VII. A GENETIC MECHANISM UNDERLYING SPECIES ISOLATION

J. T. PATTERSON AND A. B. GRIFFEN

Various types of isolating mechanisms are known to occur among different species, both in animals and in plants. Thus far, however, only a few cases have been reported in which it has been suggested that an isolating mechanism could be explained on the basis of a few simple mendelian factors. In connection with the present series of cross tests we have found a case of this character, and it is the purpose of this article to present such details as have been worked out.

In the series of tests described in the preceding article, certain cases were observed in which a disturbance in the normal sex-ratio appeared among the F_1 flies. In most of these cases the number of offspring was small and the abnormal ratio could be attributed to chance. However, there was one combination in which only males appeared in the F_1 generation from pair matings. This was the cross of *D. montana* to *D. texana*. It was found that when texana was used as the female parent, males and females were produced in about equal numbers. One count gave 172 ♀♀ and 168 ♂♂, which clearly represents a normal sex-ratio. In the reciprocal cross of montana females to texana males, only males appeared among the F_1 flies, for a total of thirty-seven (Table 5, preceding article). Although this number was small, yet the results were consistent in that these males were derived from nine different combinations involving four strains of montana and three of texana. It was therefore decided to carry out further tests with the hope of determining the nature of the mechanism involved.

EXPERIMENTAL TESTS

In a great many of the crosses between the different members of the virilis group, the number of offspring derived from pair matings is frequently small, and many of these hybrids are sterile. In order to obtain large enough numbers of hybrids for the following cross tests, it was necessary to employ mass matings. We therefore followed the plan of making up relatively large numbers of small mass matings of about twelve pairs each in vials. The number of hybrids was further increased by transferring the flies twice to fresh vials at intervals of six days. The stocks most frequently used in the cross tests were as follows: for *D. montana*, strains 1318.8a and 1218.8d, from Hart Prairie, Arizona, and Cottonwood Canyon, Utah, respectively; for *D. texana*, strains 1128.10 and 1148.9, from New Orleans, Louisiana, and Lake McKethan, Florida, respectively; for *D. americana*, a strain from Anderson, Indiana; for *D. lacicola*, strain 1360.2 from Fairbanks, Minnesota; and for *D. virilis*, the well-known Pasadena strain. Wherever other stocks have been used they are indicated in parentheses in the tables. The different forms are designated by capital letters, which represent the first letter of each specific name.

We have followed our customary procedure in indicating the origin of hybrids. Initial P_1 crosses are always recorded with the female parent as the first member of the expression; thus, T × A indicates that a texana female is crossed to an americana male. Hybrids of either sex are represented as TA. This expression indicates that the female parent of the hybrid was texana and the male parent americana. In the case of a TA male, one can see at a glance that the X chromosome is derived from texana, the Y chromosome from americana, and the autosomes from both parents.

In Table 1 we have compiled certain data which indicate the number of females and males produced in crosses of montana and lacicola to other members of the group. Females of virilis, texana, americana, and lacicola when crossed to montana males produce offspring of both sexes, and usually in normal ratios. Crosses of montana females to males of virilis, americana, and lacicola likewise give both female and male offspring, but when the male parent is texana, only male offspring are obtained. Reciprocal crosses of lacicola and texana yield hybrids of both sexes in about equal numbers.

TABLE 1

Crosses of *D. montana* and *D. lacicola* to other members of the group

Crosses	Types of matings	Females	Males
montana ♀ × virilis ♂	pairs + masses	2	3
virilis ♀ × montana ♂	pairs + masses	891	884
montana ♀ × texana ♂	pairs + masses	0	38
texana ♀ × montana ♂	pairs + masses	172	168
montana ♀ × americana ♂	pairs	42	24
americana ♀ × montana ♂	pairs	11	6
montana ♀ × lacicola ♂	pairs + masses	390	353
lacicola ♀ × montana ♂	pairs + masses	352	350
lacicola ♀ × texana ♂	masses	97	84
texana ♀ × lacicola ♂	masses	130	145

Table 2 was compiled from data obtained in both pair and mass reciprocal matings of montana females to four different types of F_1 hybrids. For each combination 100 pairs were tested for the per cent fertile. These per cents are shown in the third column. The numbers of offspring from the pair matings are supplemented by those obtained from mass matings of all excess F_1 hybrids. Hence, the number of masses employed vary, as is indicated in the parentheses in the second column. In the cross of montana females to TV males, only male offspring were produced, and in the cross of montana females to TA males, the large majority of the offspring were male. The cross of montana females to VT males also gave a majority of male offspring, but the numbers are small and the abnormal ratio may be due to chance. Finally, in the reciprocal crosses of montana to AT hybrids an excess of females was produced; this phenomenon will be discussed in a later section.

As shown in Table 1, montana females crossed to texana males produced thirty-eight males and no females. Only three strains of texana were used in the tests which gave this result. In order to determine whether all strains of texana would give this ratio consistently, montana females were crossed to six additional strains of texana males. These

TABLE 2

Four types of F_1 hybrids outcrossed to *D. montana*

Crosses	Type of matings	Per cent fertile	Females	Males
M ♀ × TV ♂	pairs + masses (16)	11	0	120
TV ♀ × M ♂	pairs + masses (12)	15	208	193
M ♀ × VT ♂	pairs + masses (18)	13	9	23
VT ♀ × M ♂	pairs + masses (12)	6	267	257
M ♀ × TA ♂	pairs + masses (6)	7	3	36
TA ♀ × M ♂	pairs + masses (9)	1	30	31
M ♀ × AT ♂	pairs + masses (72)	1	238	168
AT ♀ × M ♂	pairs + masses (66)	1	115	97

nine strains were selected from different parts of the distributional area of this species. The results obtained are given in Table 3 where the localities from which the nine strains originated are shown in parentheses.

The first five strains of texana males gave only male offspring, while the remaining four yielded both males and females. Among those producing only males, the Newton strain was by far the most fertile, giving a total of 141 F_1 males. The New Orleans and Lake McKethan strains each gave twenty males, and the Columbus strain eleven. The strain from Eva proved to be almost cross-sterile with montana, producing but two males from a single mass culture. The last four strains of texana males when mated to montana females gave some females, with the percentages varying from about seven for the Okefenokee strain to twenty for the

TABLE 3

Crosses of *D. montana* females to nine strains of *D. texana* males

Crosses	Mass matings	Masses fertile	Females	Males	Per cent males
M ♀ × T ♂ (Newton, Tex.)	29	18	0	141	100.0
M ♀ × T ♂ (New Orleans, La.)	32	5	0	20	100.0
M ♀ × T ♂ (Lake McKethan, Fla.)	41	8	0	20	100.0
M ♀ × T ♂ (Columbus, Miss.)	62	6	0	11	100.0
M ♀ × T ♂ (Eva, Tenn.)	30	1	0	2	100.0
M ♀ × T ♂ (Okefenokee Swamp, Ga.)	17	9	3	39	92.9
M ♀ × T ♂ (Morrilton, Ark.)	31	23	29	208	87.7
M ♀ × T ♂ (Hiwassee River, Tenn.)	57	9	4	22	84.6
M ♀ × T ♂ (Georgetown, Tex.)	36	13	17	68	80.0
Totals	335	92	53	531	90.9

Georgetown strain. The probable explanation for the occurrence of F_1 females in certain of these crosses will be discussed in a later section.

It is revealed in Table 2 that when montana females were crossed to TV hybrid males (Pasadena strain), only male offspring were produced. Eleven additional types of TV hybrid males have been tested to montana females. These TV hybrids carried the X chromosome of texana, the Y chromosome of virilis, and a set of autosomes of each species. The object of the experiment was to determine whether males of this type, derived from various strains of virilis, would all produce offspring that were predominantly male. The same two strains of montana (1318.8a, 1218.8d) were used as the female parent, while the TV males were obtained by using females of the New Orleans and Lake McKethan strains of texana and virilis males from the twelve different strains. The strains of virilis originated from flies which had been collected at widely scattered places; two from Manchukuo, one from Japan, one from Mexico, and eight from various points in the United States (Table 4).

Table 4

Crosses of *D. montana* females to *texana/virilis* males and *texana/lacicola* males

Crosses	Mass matings	Masses fertile	Females	Males	Per cent males
M ♀ × TV ♂ (Shenking, Manchukuo)	45	33	0	184	100.0
M ♀ × TV ♂ (Pasadena, from N.Y.)	28	17	0	120	100.0
M ♀ × TV ♂ (Tokyo, Japan)	45	28	0	115	100.0
M ♀ × TV ♂ (Mexico City, Mexico)	45	33	1	276	99.6
M ♀ × TV ♂ (New Orleans, La.)	45	39	2	198	99.0
M ♀ × TV ♂ (Galveston, Texas)	45	18	1	102	99.0
M ♀ × TV ♂ (Brooksville, Fla.)	45	43	3	279	98.9
M ♀ × TV ♂ (Henly, Texas)	38	17	1	76	98.7
M ♀ × TV ♂ (Memphis, Tenn.)	45	35	2	153	98.7
M ♀ × TV ♂ (San Antonio, Texas)	45	31	2	138	98.5
M ♀ × TV ♂ (Beaumont, Texas)	45	29	2	129	98.4
M ♀ × TV ♂ (Mukden, Manchukuo)	42	25	2	126	98.4
Totals	513	348	16	1,896	99.2
M ♀ × TL ♂ (Fairbanks, Minn.)	12	11	0	68	100.0

The first three crosses listed in the table (Shenking, Pasadena, Tokyo) produced only male offspring, while each of the other nine gave one or more females in addition to a large number of males. The total number of offspring obtained in this series of tests was 1,912, of which 1,896, or 99.2%, were males. About 120 of these males were mated to different types of females, but no offspring was produced. It is therefore very probable that all of them were sterile. Of the sixteen females listed in the table, seven died within a few hours after emerging from the puparia. Each of the other nine was crossed to at least two different types of males. Eight of these proved to be sterile, and one (Memphis strain) gave a few larvae which failed to develop as far as the pupa stage.

In the last line of this table are shown the results derived from the cross of montana females to texana/lacicola hybrid males (TL). The eleven fertile mass cultures produced 68 males and no females. This result is in line with those obtained in the TV crosses, and together they demonstrate that the effect of the X chromosome of texana is not modified by the other chromosomes of either virilis or lacicola.

In Table 5 are listed the results from tests of various types of F_1 hybrids to members of the virilis group. In the first set of tests ML hybrids were crossed to one strain of virilis, two of texana, and one of americana. Only one mating proved to be incompatible (ML ♀ × V ♂); all others gave approximately equal numbers of males and females, except one. The exception was the cross of ML ♀ × A ♂, where the number of females was greatly in excess of the number of males (86 vs. 35). This result is similar to the one recorded in Table 2 between montana females and AT hybrid males.

In the second set of tests LM hybrids were crossed to virilis, texana, and americana. Here again, only one mating proved to be incompatible (V ♀ × LM ♂). The other five crosses gave both males and females, although in two of them the numbers of offspring were small.

In the third set of tests VM hybrids were tested to two strains of texana. Both matings of VM males failed to give offspring, but the reciprocal crosses yielded males and females in about equal numbers.

TABLE 5

Crosses between virilis, texana, and americana and various types of F_1 hybrids

Crosses	Masses mated	Masses fertile	Females	Males
ML ♀ × V ♂ (Pasadena)	40	0	------	------
V ♀ × ML ♂	40	26	297	302
ML ♀ × T ♂ (New Orleans)	40	1	46	50
T ♀ × ML ♂	40	2	19	18
ML ♀ × T ♂ (Lake McKethan)	40	6	67	62
T ♀ × ML ♂	28	2	19	16
ML ♀ × A ♂ (Anderson)	40	11	86	35
A ♀ × ML ♂	36	2	24	22
LM ♀ × V ♂ (Pasadena)	13	1	8	10
V ♀ × LM ♂	20	0	------	------
LM ♀ × T ♂ (New Orleans)	20	1	1	2
T ♀ × LM ♂	20	7	73	54
LM ♀ × A ♂ (Anderson)	20	2	2	3
A ♀ × LM ♂	20	10	151	117
VM ♀ × T ♂ (New Orleans)	28	9	12	13
T ♀ × VM ♂	33	0	------	------
VM ♀ × T ♂ (Lake McKethan)	19	9	19	24
T ♀ × VM ♂	19	0	------	------
TM ♀ × T ♂ (New Orleans)	10	3	6	6
M ♀ × TM ♂	10	0	------	------
M ♀ × TV^G ♂	41	14	0	41+, 14^G
TV^G ♀ × M ♂	39	26	39+, 24^G	31+, 22^G

In the fourth set of tests TM hybrids were crossed to texana. In the cross, TM ♀ × T ♂, the males and females appeared in equal numbers, but the reciprocal mating was incompatible.

Finally, in the last set of tests reciprocal crosses of TV^G hybrids to montana were carried out. The first cross, M ♀ × TV^G ♂, gave 41 plus and 14 Garnet males, but no females. The reciprocal cross produced 39 plus and 24 Garnet females, and 31 plus and 22 Garnet males. In both crosses the number of Garnet flies was less than the number of plus flies, indicating that the former class is less viable than the latter. Since chromosomes 3 and 4 of texana are fused, the results show that these two autosomes do not modify the effect of the texana X chromosome in eliminating the female zygotes.

The results from the cross tests listed in Tables 3–5 show that there is present in the X chromosome of texana a factor (or factors) which brings about the elimination of all but a few of the hybrid females when introduced into the egg of montana by the sperm. In this connection, it is well to remember that there are other factors which greatly reduce the number of offspring produced in interspecific crosses among members of the virilis group. The low fertility rate in many such crosses is due to a strong sexual isolation, or a failure of the flies to mate. We have also shown (Patterson, Stone, and Griffen, 1942) that in addition to sexual isolation, many of the male gametes are inactivated or killed in the reproductive tract of the female after cross mating has taken place. Thus in the virilis/americana cross only 15% of the eggs laid during the first day following mating were found to be inseminated, and only 3% of the eggs kept for development produced offspring. Eggs laid after the first day never developed. Hence, if a female is to continue to lay fertile eggs, she must be remated at least once in every twenty-four hours.

The inactivation or death of the sperm in such cases is due, apparently, to an incompatibility between the foreign sperm and the secretions of the female ducts. But neither this effect nor that of sexual isolation will explain the results obtained in the montana/texana cross, because those effects are not differential. In this case there is an additional mechanism the effect of which is either to eliminate the X-bearing sperm, or else to cause the death of the female zygote during the course of development.

In order to determine which of these two possible explanations might be the correct one, we carried out an experiment similar to the one on the virilis/americana cross referred to above. The females of montana were crossed to two types of TV males. In the first cross TV (Henly) males were used (Table 6). A total of 680 eggs were checked by the smear method for the presence of sperm, and 67, or 9.8%, were found to have been inseminated. Of the 833 eggs laid by the same females and kept for possible development, none produced adults, although a few small larvae appeared in the cultures. In the second cross TV (Mukden) males were used, and 14 eggs out of 198 checked contained sperm, for

TABLE 6

Inseminated eggs and egg hatch in crosses of *D. montana* ♀♀ to TV ♂♂

Crosses	Eggs checked for sperm	Number with sperm	Eggs kept for development	Number of adult flies
M ♀ × TV ♂ (Henly)	680	67	833	larvae but no adults
M ♀ × TV ♂ (Mukden)	198	14	344	4 males

a percentage of 7.0. Of the 344 eggs kept for development, four produced pupae from which males emerged, for a percentage of 1.1.

These results are somewhat similar to those obtained in the virilis/americana test. In both cases the per cent of hatch is very much lower than the per cent of eggs containing sperm, showing that not all inseminated eggs produce adults. If these two per cents had been equal in the montana/TV test, it would have demonstrated that the differential mortality had occurred among the male gametes. As it is, it seems more probable that the female zygotes die off during the course of development. There are some observations which support this conclusion. In the experiment in which montana females were crossed to various types of TV males, some dead larvae of various sizes were observed in the cultures. A few of these had reached full growth at the time of death, and one large moribund specimen was found to be a female. Moreover, several dead females were found in pupae from which adults had failed to emerge.

DISCUSSION AND CONCLUSIONS

Since this article is preliminary in character we will not attempt to give a full discussion of the phenomenon reported above. There are various experiments now in progress which must be completed before a final analysis of the case can be made. In the meantime, it is possible to offer some interpretations of the facts already established.

It is clear from the experimental data that there is a sex-linked factor in *D. texana* which is not effective within its own species. Consequently, it will not be selected against, and will not be eliminated through natural selection. There is also a factor in *D. montana* which is non-effective within its own species, and it will likewise not be selected against. But when these two factors are brought together, through the fertilization of the montana egg by the X-bearing sperm of texana, they interact and produce inviable female zygotes which nearly all die during development. The few F_1 females which do survive are weak and apparently are not able to produce viable offspring, while the F_1 males are sterile. This case represents an isolating mechanism which tends to prevent an exchange of genes in certain interspecific crosses. Three somewhat similar cases have been reported in the literature, one in Drosophila and two in plants.

In 1942 Dr. J. F. Crow reported from this laboratory a case involving crosses between *D. mulleri* and certain strains of *D. aldrichi,* designated as aldrichi-2. He found that when aldrichi-2 males were crossed to mulleri females the offspring were predominantly male. He suggested that this result could be explained by assuming the presence of a gene (or genes) on the X chromosome of aldrichi-2, which, while having no noticeable effect within the species, acts as a dominant semilethal in interspecific crosses. This case differs from the one here under discussion in that the reciprocal mating is incompatible, whereas in our case it is cross-fertile and produces males and females in equal numbers.

One of the cases in plants was first reported by Babcock and Collins (1920a, 1920b), who described hybrids between *Crepis capillaris* and *C. tectorum* which died after developing to the cotyledon stage. The case was further studied in detail by Hollingshead (1926), who found that the lethal responsible for the production of inviable hybrids was present in certain strains of *C. tectorum,* and showed that the genotype of the *tectorum* plant could be either homozygous for the lethal, heterozygous for it and its allele, or homozygous for its alleles. She was not able to determine a dominant or recessive relationship, because this was precluded by the fact that the lethal has no effect when present doubly in *tectorum* and exhibits its characteristic effect when present singly in hybrids. Evidence was found that this lethal was effective in hybrids between *tectorum* and two other species of *Crepis,* but non-effective in hybrids with a third and fourth species.

The second case in plants is the "crumpled" character in cotton, a condition that was first observed by Harland (1915). However, the genetics underlying this character was not understood until seventeen years later, when Hutchinson (1932) published an account of his studies on this abnormal sterile or semi-sterile type (crumpled) which appeared in progeny of a cross between wild cotton of the Sudan, *Gossypium Nanking* var. *soudanensis* and a strain of *G. arboreum* var. *sanguinea.* He studied the behavior of the factors in considerable detail, because, as he states, cases of interspecific sterility that are capable of exact genetic analysis are seldom found. He found that the crumpled condition, which expresses itself in varying degrees in phenotype and in sterility, was due to a pair of complementary factors which he designated by the letters A and B. The factor A was found to be present in two strains of *G. Nanking* var. *soudanensis,* while the factor B was present in seventeen varieties of *G. arboreum, G. Nanking, G. hebaceum,* and *G. obtusifolium,* and in a strain of *Stocksii* from Sind. He also demonstrated the existence of a number of modifying factors affecting the degree of expression of the crumpled character.

More recently, Silow (1941) has given much additional information on the crumpled character. In the meantime, the complementary factors A and B of Hutchinson have had assigned the symbols Cp_a and Cp_b, respectively (Hutchinson and Silow, 1939). The Cp_a gene was found in one strain of *arboreum,* and the Cp_b gene in 25 of 41 tested strains of *ar-*

boreum and *herbaceum.* In *Gossypium* this complementary gene system is effective within a single species, as well as in interspecific crosses.

These four cases, while differing from one another, show several points of similarity. At least three of them can be explained on the basis of a complementary gene system accompanied by modifying factors and sterility. The important point is whether such a system plays any rôle in species isolation. Neither Hollingshead nor Silow believes that the system has played much part in the late evolutionary history of the species investigated. The comments of these two authors are well worth quoting.

Hollingshead (1926, pp. 137–138) makes the following statement: "From an evolutionary standpoint there is no evidence as to whether this Crepis lethal or its allelomorph is the older. Both were found in Western European and Central Siberian strains, indicating a wide range for each. Since the lethal has no effect on the parental species, and since normal F_1 interspecific hybrids are almost sterile it is difficult to see how it has had any part in the later evolutionary development of the species investigated. If the lethal is the older its present existence may be a relic from a time when the species concerned were in the process of differentiation and it may have played a rôle in the maintenance of the ancestral stocks."

Silow's remarks (1941, p. 352) are as follows: "The system may have been of importance as an isolation barrier at some past time, and might easily become so again under particular local conditions, but, as in *Crepis,* it does not appear to have any great significance at the present time. The whole situation in *Gossypium* points to a lack of harmony as the fundamental cause of interspecific incompatibility in this genus as it exists now, and, as has been shown, geographic separation appears to be a potent influence in the accumulation of many small differences into a clear distinction."

While we are unable at present to assign any past evolutionary significance in the montana-texana case, yet it can be shown that the complementary gene system is an effective isolating mechanism in these forms. This mechanism operates as a selective agency which makes it difficult for the X-bearing sperm of texana to produce viable zygotes with montana eggs. From this fact, which is clearly shown in the tables, it is evident that the montana factor (or factors) finds its incompatibility complement in the X chromosome of texana. The eggs of texana are successfully fertilized by both X-bearing and Y-bearing sperm of montana and produce male and female sterile hybrids in equal numbers. On the other hand, the eggs of montana, although fertilized successfully by both the X-bearing and Y-bearing sperm of texana, produce non-viable female zygotes and sterile male hybrids. Thus, the mechanism consists in the incompatibility of the montana egg protoplasm with a factor (or factors) which is confined to the texana X chromosome.

The absence of complementary autosomal factors in texana is clearly shown in the crosses M ♀ × TV ♂ and M ♀ × VT ♂ (Table 4). In the first instance, the sperm carried the texana X accompanied by autosomes which were T, V, or segregants of T and V, and the offspring were predominantly male with less than one per cent of poorly viable females. In the second instance (M ♀ × VT ♂), the sperm carried the virilis X, accompanied by autosomes which were V, T, or segregants of V and T. In this cross males and females were produced in equal numbers. Therefore, the texana X-factor is independent of modifiers or complements in the texana autosomes as already indicated in the TV^G test, and similarly is independent of virilis autosomal factors. Comparable results were observed in the cross of M ♀ × TL ♂ (Table 4).

Having demonstrated that the texana factors are restricted to the X chromosome, we then tested for the number and location of these factors, as follows: The cross V ♀ × T ♂ was made, using a virilis stock in which the X chromosome bore the markers y^{40a} *ec cv v* si^2 *dy* ap^{40e} (y^{40a}*–ple*). The F_1 females were crossed to fresh y^{40a}*–ple* males and F_2 crossover and non-crossover males were selected. Repeated backcrosses to fresh y^{40a}*–ple*, carried through the F_6 generation, yielded stocks with several X chromosome crossover types accompanied by virilis autosomes. Males of these stocks were outcrossed to montana females. Only male offspring were produced when the recombination X chromosomes carried the texana equivalent of the *ec*–si^2 interval; all other recombinations produced offspring of both sexes. Therefore, the texana X-factor (or factors) is confined to this region of the chromosome. Further tests that were carried out yielded only a few offspring, but these indicated that the *ec–cv* interval represents the effective segment, which in turn suggests that a single factor is probably involved. In any case, we see no reason for assuming that more than one factor is present in this short segment.

The results of the crosses, M ♀ × A ♂, M ♀ × TA ♂, and M ♀ × AT ♂ (Tables 1 and 2), present an interesting situation. The cross M ♀ × TA ♂ yielded three females and thirty-six males, a ratio which indicates the presence of an americana autosomal factor modifying the texana X-factor. The autosomal segregation possibilities of TA hybrids limit the factors to chromosome 5 of americana. In the crosses M ♀ × A ♂ and M ♀ × AT ♂ the preponderance of female offspring may indicate the absence of the T-factor in the americana X chromosome, or its suppression by the autosomal modifier; in either case we should expect a normal sex ratio among the offspring. The excess of female offspring can be explained only as an expression of the traditional low viability of the hemizygous sex in interspecific hybrids. Crossover and segregation tests of the americana X and autosomes are in progress, since it is necessary to determine whether americana possesses the T-factor. It will be recalled that the americana X was shown to have been derived originally from that of texana at the time of the hybrid origin of americana. If the present complementary factor of texana was present at the time of the americana origin, this factor should be present in the several stocks of americana. A further point of interest is that several texana stocks, as indicated in Table 3, produce a small proportion of female hybrids in crosses to montana. These strains were derived from localities which, with one exception, lie within the overlapping zone of amaricana and texana populations; hence, the americana autosomal modifier may be present in these texana strains as a result of recent contacts between the two species.

By the way of summary, the texana phase of the isolating mechanism consists of an X-chromosomal factor located in the interval *ec–si²* of the chromosome. This factor is independent of others in the X chromosome, and in the autosomes not only of texana but of all other tested species of the virilis group, with the exception that an autosomal suppressor is indicated in americana. Inasmuch as the texana factor operates in the heterozygote it may be considered in effect a dominant; yet this classification cannot be assigned since its expression depends upon the previous operation of the montana factors. Experiments are under way to determine the effects of the T-factor in the homozygous condition both within its own species and in montana-texana hybrids.

The montana phase of the isolating mechanism consists of factors (or a factor) which are simple recessives. As shown in the tables, TM, MV, ML, and the reciprocals of the last two produce equal numbers of male and female offspring when crossed to texana males. The MT females represent the inviable class of zygotes, and AM and MA females have thus far proved sterile; therefore these two classes are not available for testing. It is clear that the montana factors are recessive to those of V, L, and T and therefore achieve expression only when homozygous. A further consideration is that, as shown in the cross TM ♀ × T ♂, neither the M-factors nor the T-factor functions in the protoplasm of the hybrid female. We must conclude that the M-factors express themselves only in the laying down of the egg protoplasm and that the T-factor comes into expression only when the montana eggs are fertilized. Thus the M-factors act previous to fertilization by setting up protoplasmic characteristics which provide an effective selection mechanism against the X-bearing sperm of texana. An effort was also made to determine the location of the montana factors concerned, but the necessary hybrids with which to make this determination proved sterile.

REFERENCES

Babcock, E. B., and Collins, J. L., 1920a. Interspecific hybrids in Crepis. I. *Crepis capillaris* × *C. tectorum*. Univ. Calif. Pub. Agric. Sci., 2:191–204.

————, 1920b. Interspecific hybrids in Crepis. Proc. Nat. Acad. Sci. Wash., 6:670–673.

Crow, J. F., 1942. Cross fertility and isolating mechanisms in the *Drosophila mulleri* group. Univ. of Texas Publ., 4228:53–67.

Harland, S. C., 1915. Report of the Agricultural Experiment Station in St. Croix, 1913–14. Appendix.

Hollingshead, Lillian, 1930. A lethal factor in Crepis effective only in an interspecific hybrid. Genetics, 15:114–140.

Hutchinson, J. B., 1932. The genetics of cotton. Part VII. "Crumpled": A new dominant in Asiatic cottons produced by complementary factors. Jour. of Genetics, 25:281–291.

Hutchinson, J. B., and Silow, R. A., 1939. Gene symbols for use in cotton genetics. Jour. of Heredity, 30:461.

Silow, R. A., 1941. The comparative genetics of *Gossypium anomalum* and the cultivated Asiatic cottons. Jour. of Genetics, 42:259–358.